apify

Enabling agents to pay: **Implementation challenges of x402 in production**

Stepan Skopek, Engineer
@stepskop

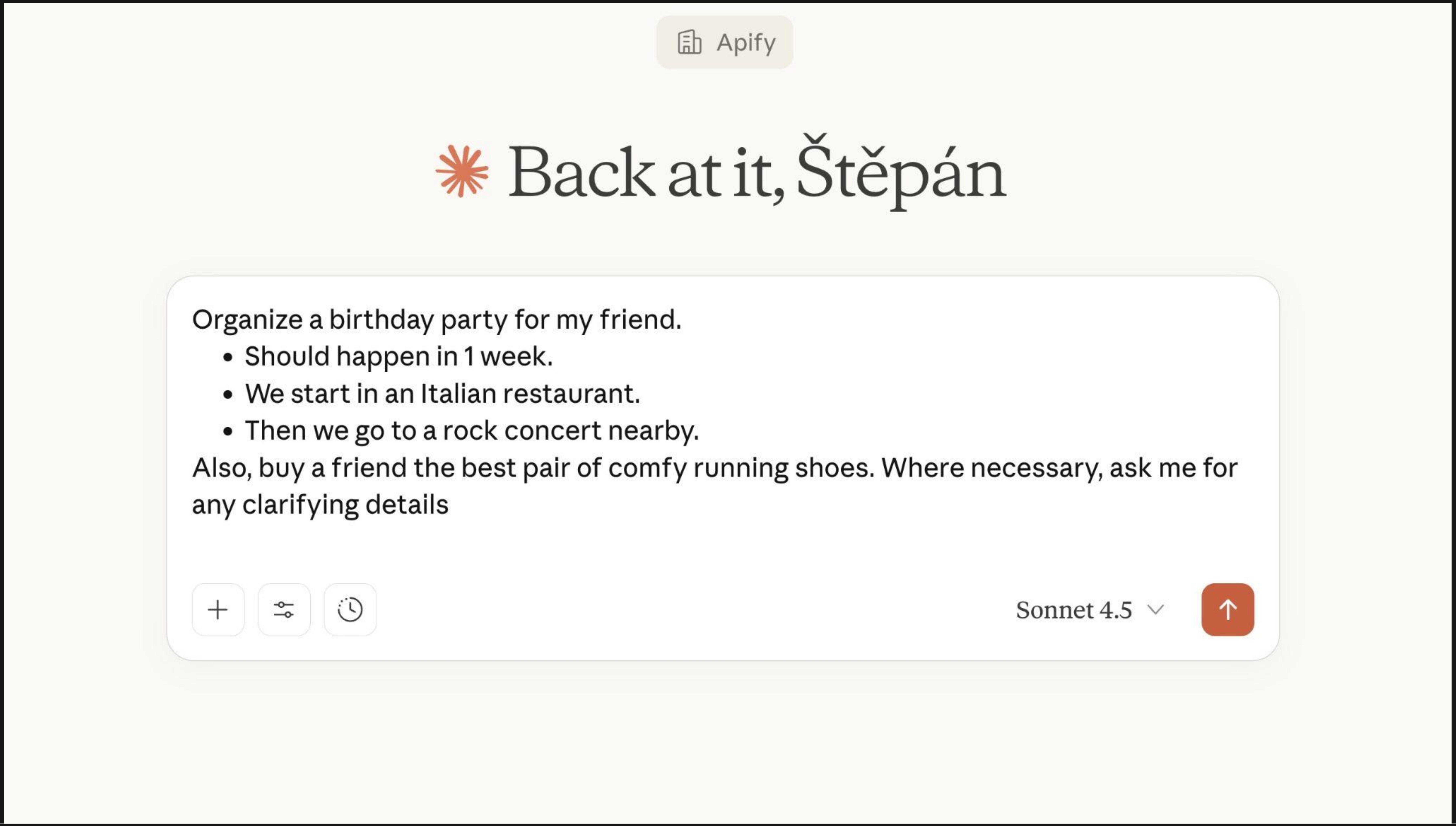
AI AGENTS

goal-oriented systems

that make

independent decisions

Using **AI agents**



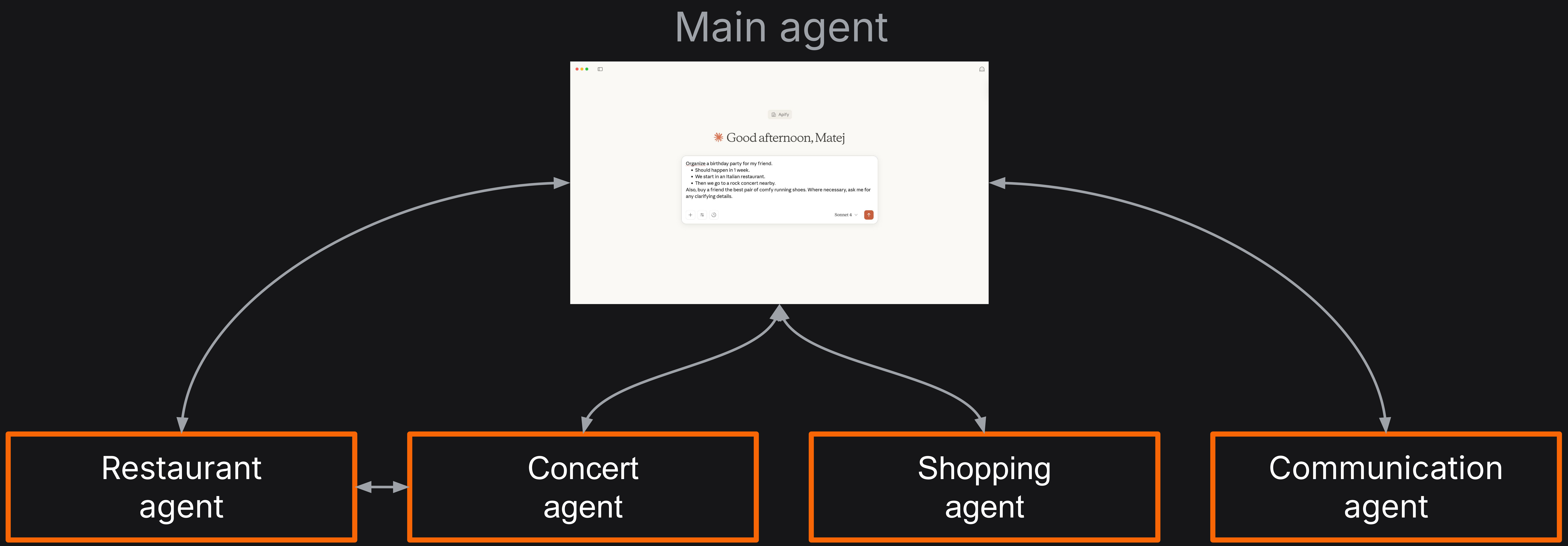
Back at it, Štěpán

Organize a birthday party for my friend.

- Should happen in 1 week.
- We start in an Italian restaurant.
- Then we go to a rock concert nearby.

Also, buy a friend the best pair of comfy running shoes. Where necessary, ask me for any clarifying details

Behind the scenes



Main agent

Apify

Good afternoon, Matej

Organize a birthday party for my friend.
- Should happen in 1 week.
- We start in an Italian restaurant.
- Then we go to a rock concert nearby.

Also, buy a friend the best pair of comfy running shoes. Where necessary, ask me for any clarifying details.

Restaurant agent

Concert agent

Shopping agent

Communication agent

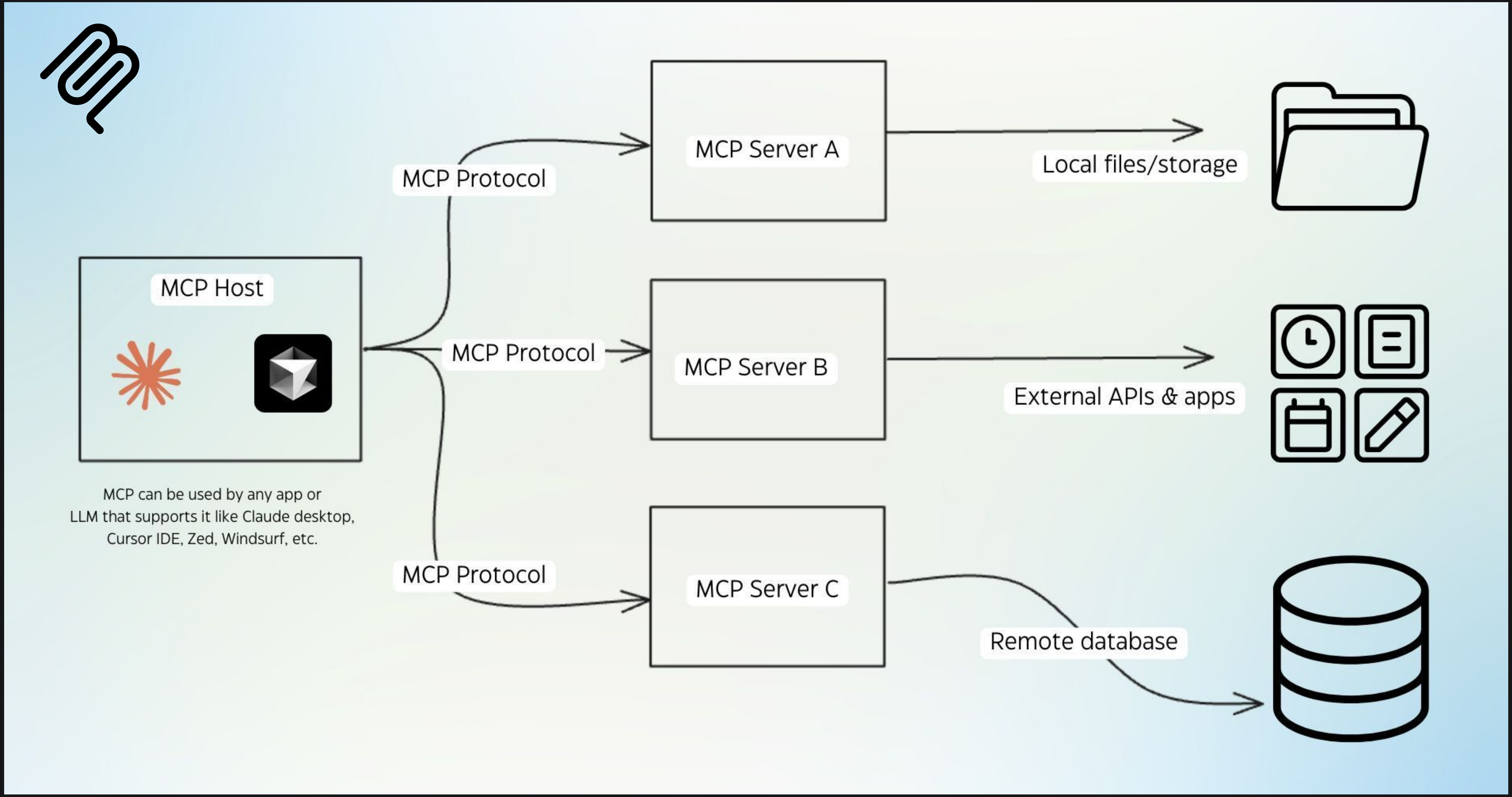
How do the **agents** **act in real world**?

Tools

interfaces to the world

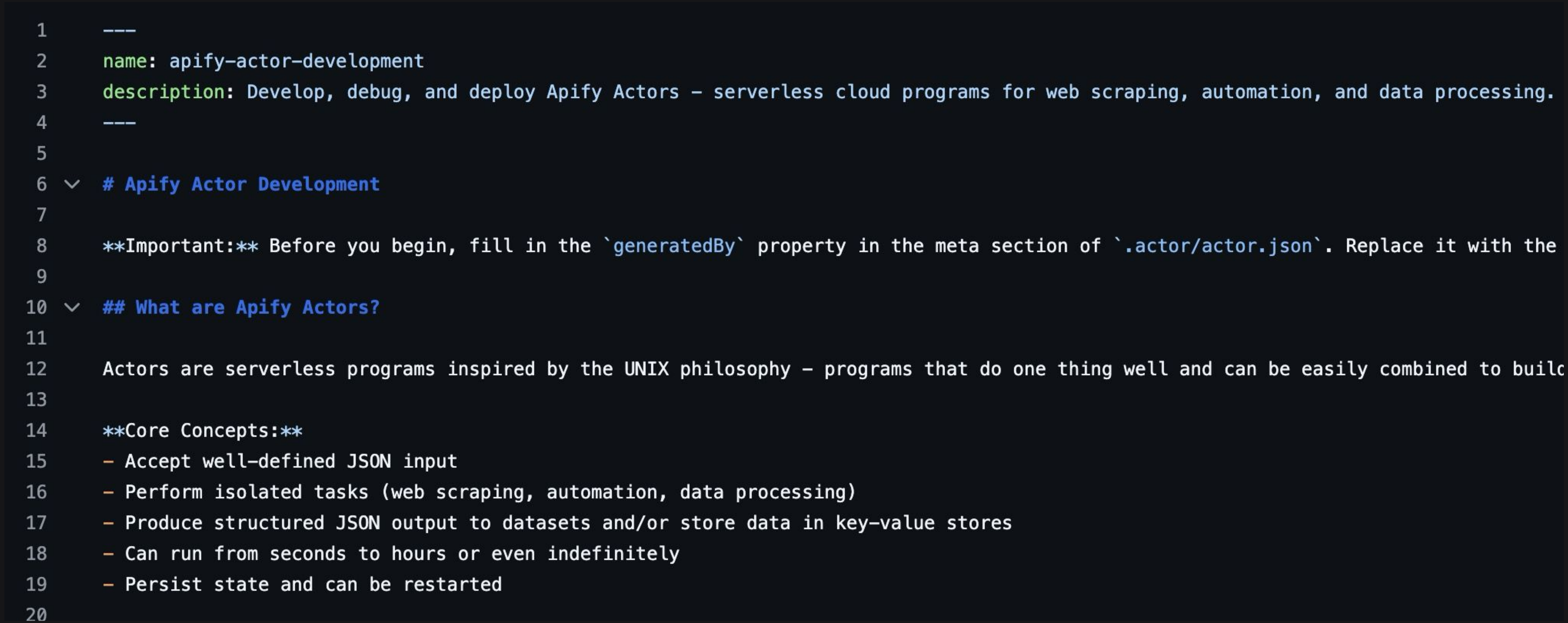
How **agents** use **tools**?

Model Context Protocol

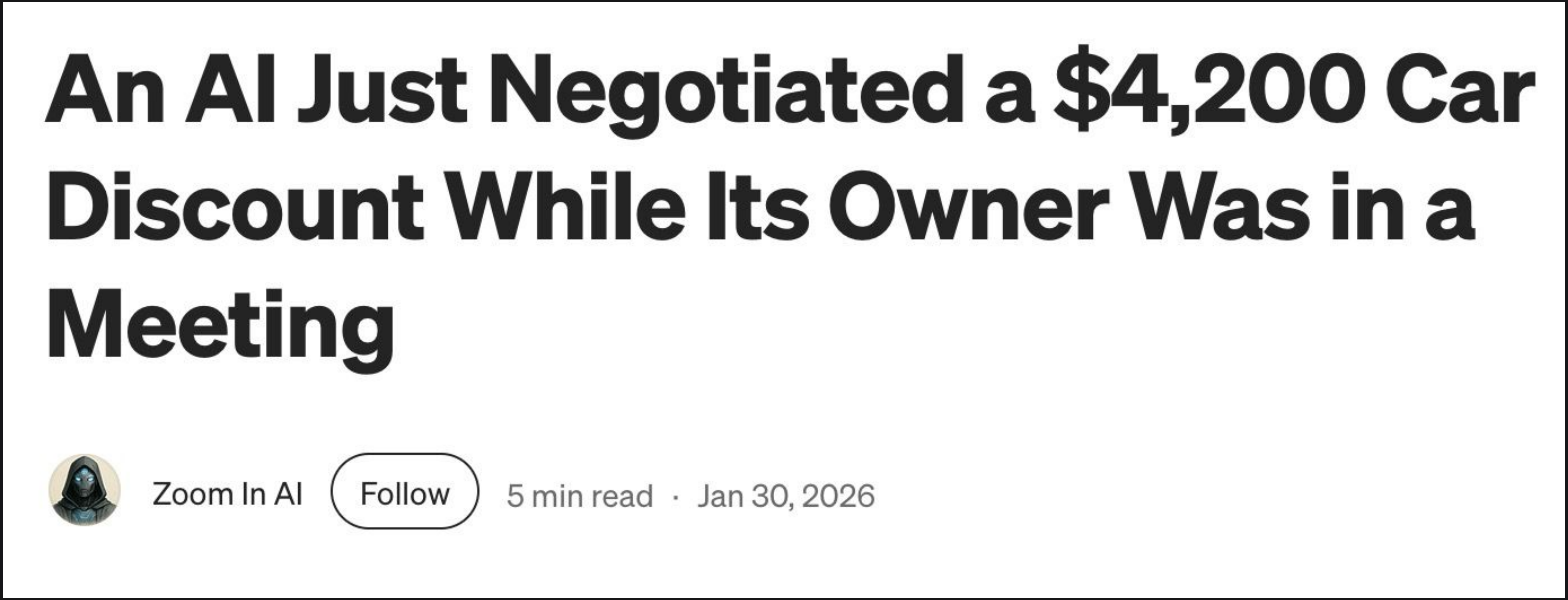

MCP Host
MCP Protocol
MCP Protocol
MCP Protocol
MCP Server A
MCP Server B
MCP Server C
Local files/storage
External APIs & apps
Remote database

MCP can be used by any app or
LLM that supports it like Claude desktop,
Cursor IDE, Zed, Windsurf, etc.

Agent **Skills**



```
---
name: apify-actor-development
description: Develop, debug, and deploy Apify Actors - serverless cloud programs for web scraping, automation, and data processing.
---

# Apify Actor Development

**Important:** Before you begin, fill in the `generatedBy` property in the meta section of `.actor/actor.json`. Replace it with the

## What are Apify Actors?

Actors are serverless programs inspired by the UNIX philosophy - programs that do one thing well and can be easily combined to buil

**Core Concepts:**
- Accept well-defined JSON input
- Perform isolated tasks (web scraping, automation, data processing)
- Produce structured JSON output to datasets and/or store data in key-value stores
- Can run from seconds to hours or even indefinitely
- Persist state and can be restarted

```



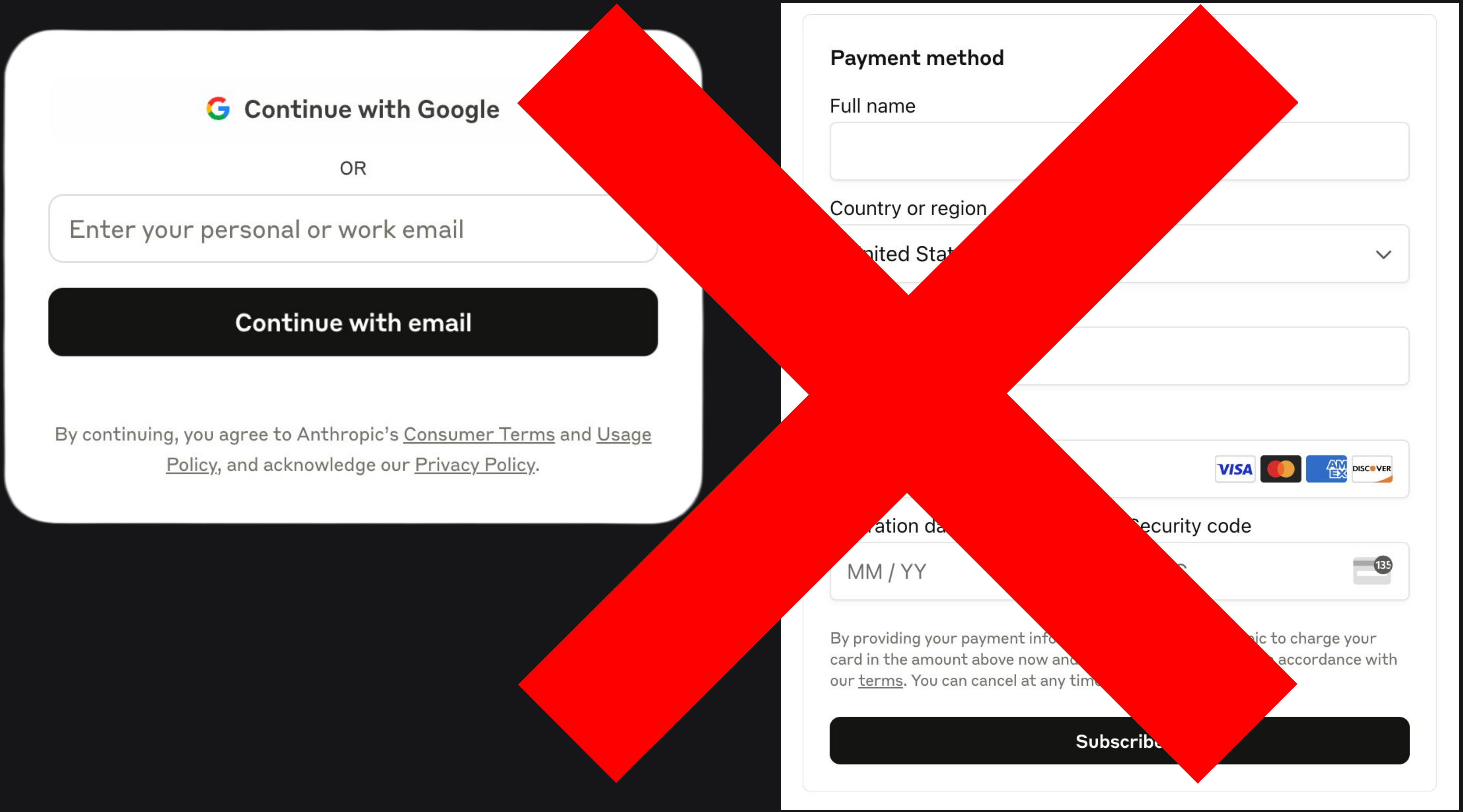
An AI Just Negotiated a $4,200 Car Discount While Its Owner Was in a Meeting

Zoom In AI

5 min read · Jan 30, 2026

How do **agents** **pay**?

Give **agent** your **credit card**?



Continue with Google

OR

Enter your personal or work email

Continue with email

By continuing, you agree to Anthropic's Consumer Terms and Usage Policy, and acknowledge our Privacy Policy.

Payment method

Full name

Country or region

nited Stat

ation da

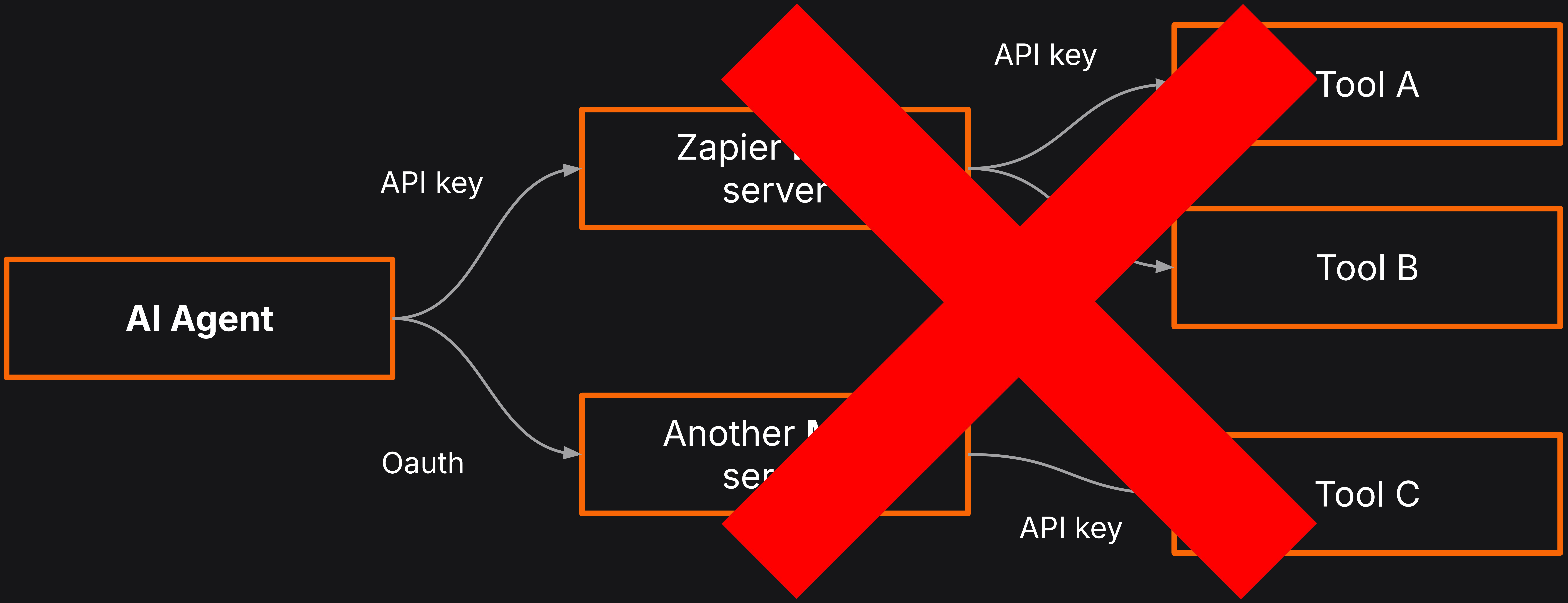
ecurity code

MM / YY

By providing your payment info ic to charge your card in the amount above now and accordance with our terms. You can cancel at any tim

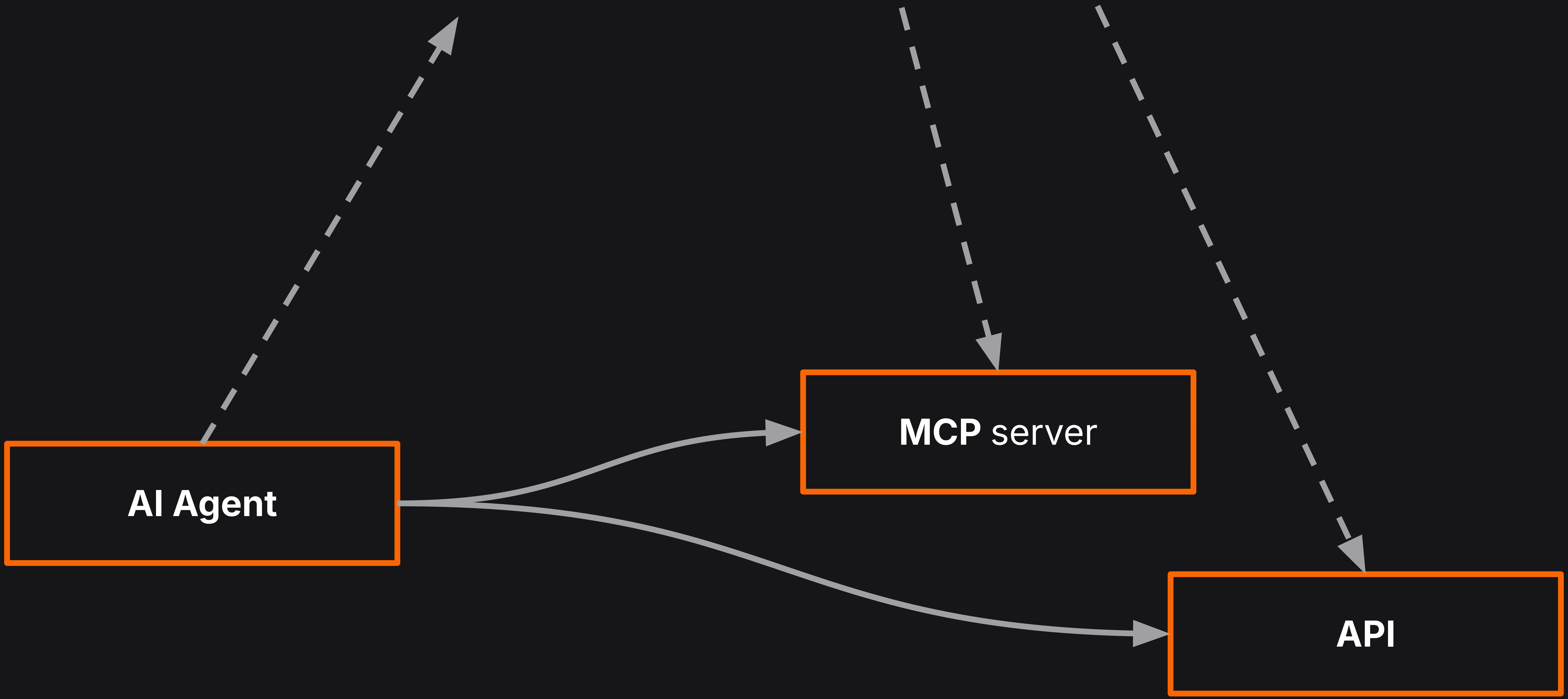
Subscrib

Subscribe to each service manually?


API key
API key
Zapier
server
Tool A
Tool B
AI Agent
Another
ser
Oauth
API key
Tool C

Agentic **payment protocols** & **providers**

Standard for **payments** & **identity**



AI Agent

MCP server

API

L402

MPP

KYAPay ✅

ACP

x402

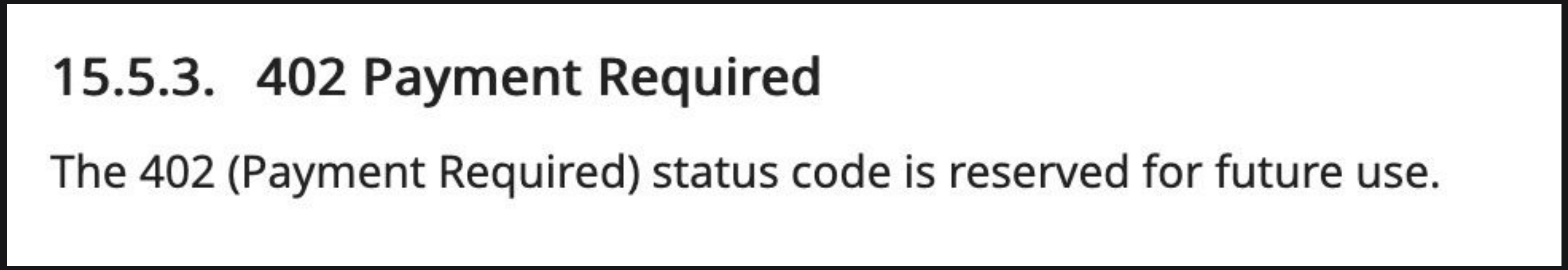
UCP

A2P

Visa Intelligent Commerce

x402 V2

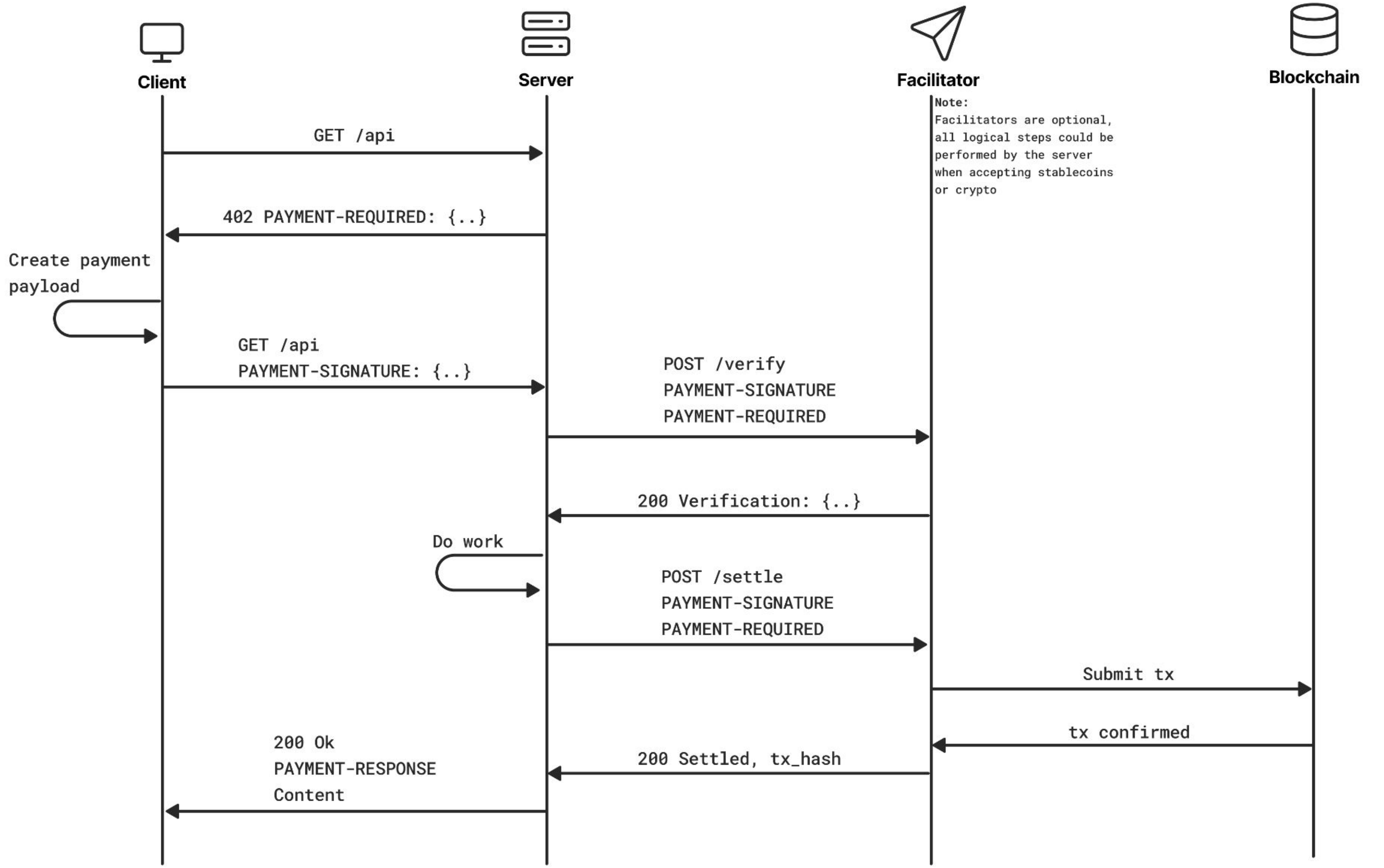
- open source payment protocol by Coinbase
- HTTP, 402 status code
- request/response headers-only
- built on blockchain (multiple networks)
- decentralized



15.5.3. 402 Payment Required

The 402 (Payment Required) status code is reserved for future use.

x402 V2 flow



Client

Server

Facilitator

Blockchain

Note:
Facilitators are optional,
all logical steps could be
performed by the server
when accepting stablecoins
or crypto

GET /api

402 PAYMENT-REQUIRED: {..}

Create payment
payload

GET /api
PAYMENT-SIGNATURE: {..}

POST /verify
PAYMENT-SIGNATURE
PAYMENT-REQUIRED

200 Verification: {..}

Do work

POST /settle
PAYMENT-SIGNATURE
PAYMENT-REQUIRED

Submit tx

tx confirmed

200 Settled, tx_hash

200 Ok
PAYMENT-RESPONSE
Content

Full **x402** tech spec

github.com/x402-foundation/x402

Implementing **x402** into existing systems

Payment scheme

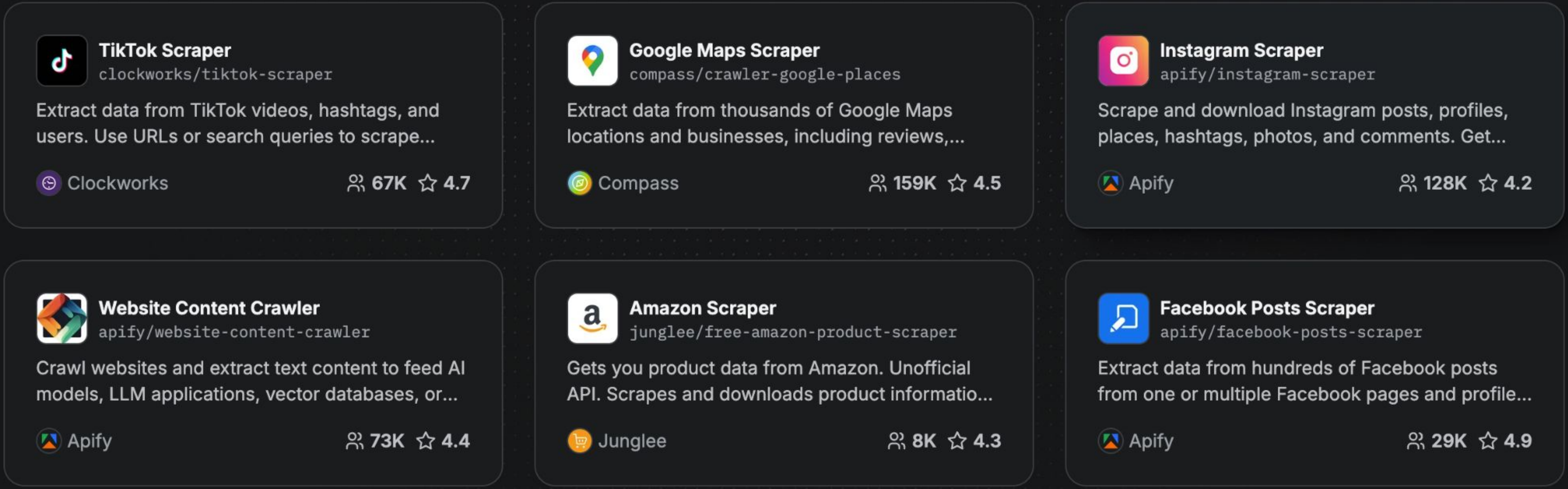
exact scheme

- first and only supported scheme
- server declares an *exact* cost upfront, client pays precisely that amount
- fixed-cost API calls, static resources

…won't work well for:

- LLM token generation
- querying a database
- **web scraping**

Apify runs a marketplace of **24,000+ tools** we call **Actors**



TikTok Scraper
clockworks/tiktok-scraper
Extract data from TikTok videos, hashtags, and users. Use URLs or search queries to scrape...
Clockworks 67K ☆ 4.7

Google Maps Scraper
compass/crawler-google-places
Extract data from thousands of Google Maps locations and businesses, including reviews,...
Compass 159K ☆ 4.5

Instagram Scraper
apify/instagram-scraper
Scrape and download Instagram posts, profiles, places, hashtags, photos, and comments. Get...
Apify 128K ☆ 4.2

Website Content Crawler
apify/website-content-crawler
Crawl websites and extract text content to feed AI models, LLM applications, vector databases, or...
Apify 73K ☆ 4.4

Amazon Scraper
junglee/free-amazon-product-scraper
Gets you product data from Amazon. Unofficial API. Scrapes and downloads product informatio...
Junglee 8K ☆ 4.3

Facebook Posts Scraper
apify/facebook-posts-scraper
Extract data from hundreds of Facebook posts from one or multiple Facebook pages and profile...
Apify 29K ☆ 4.9

A: Wait for ***upto*** scheme

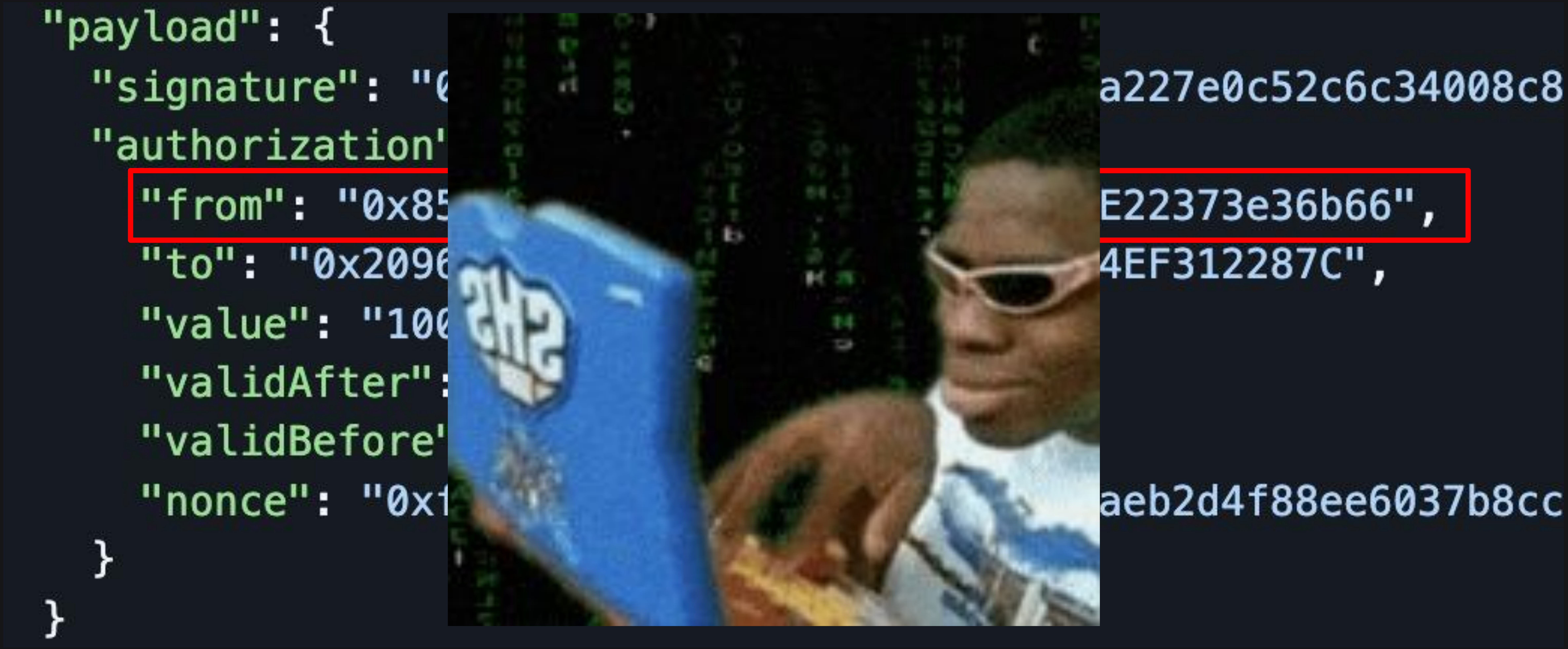
B: Adapt your pricing model

C: Hack it

C: **exact** scheme with **refunds**

- accept the *exact* payment with **estimated price, upper bound**
- internal bookkeeping - keep track of the usage
- after the job is done, **return unused funds manually, on blockchain**

Initial payment payload from the agent:



```
"payload": {
  "signature": "0                          a227e0c52c6c34008c8
  "authorization"
    "from": "0x85                        E22373e36b66",
    "to": "0x2096                        4EF312287C",
    "value": "100
    "validAfter":
    "validBefore"
    "nonce": "0xf                        aeb2d4f88ee6037b8cc
  }
}
```

Apify is on **x402**!

Available on both **API** & **MCP**!

docs.apify.com/platform/integrations/x402

MCP and x402

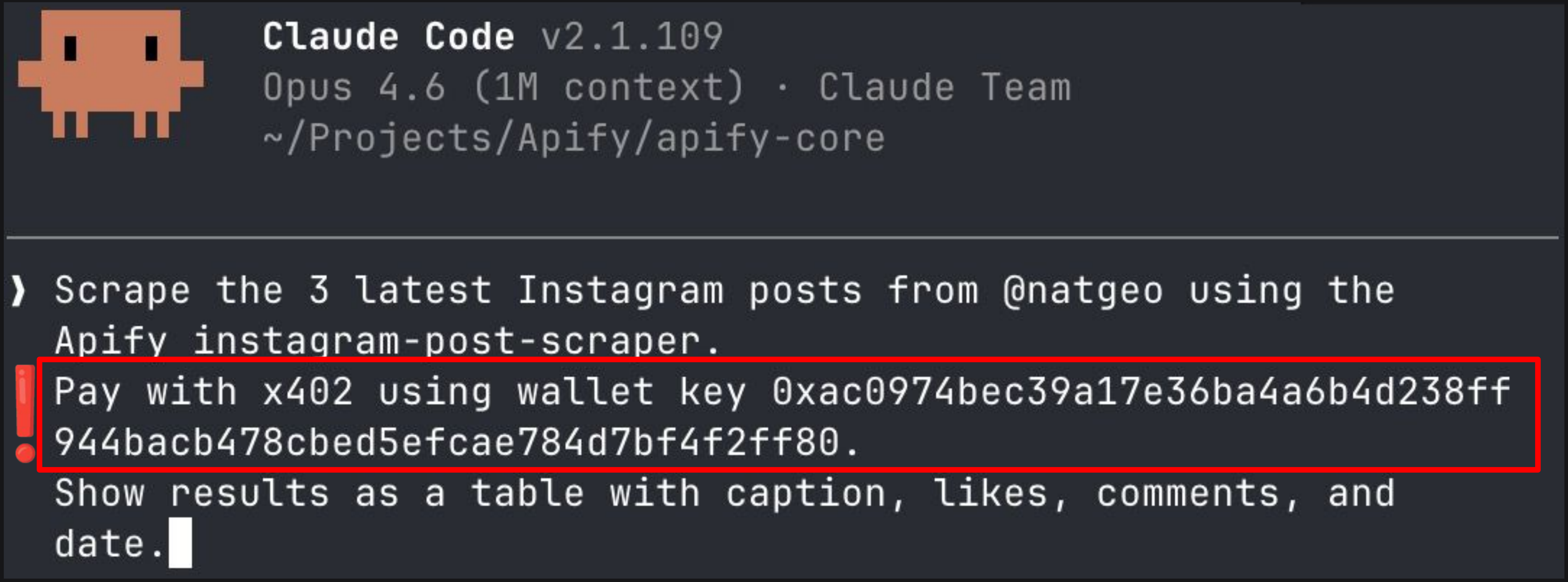
Connect to MCP server in `x402 mode`

mcp.apify.com?payment=x402

How to connect your **wallet**?

Give your agent the **wallet private key** 😱

- agent will create the signing code and execute it



```
Claude Code v2.1.109
Opus 4.6 (1M context) · Claude Team
~/Projects/Apify/apify-core

❯ Scrape the 3 latest Instagram posts from @natgeo using the
  Apify instagram-post-scraper.
  Pay with x402 using wallet key 0xac0974bec39a17e36ba4a6b4d238ff
  944bacb478cbed5efcae784d7bf4f2ff80.
  Show results as a table with caption, likes, comments, and
  date.
```

Say hello to **mcpc** 👋
- the universal CLI client for MCP

```
npm install -g @apify/mcpc
```

Supports agentic payments

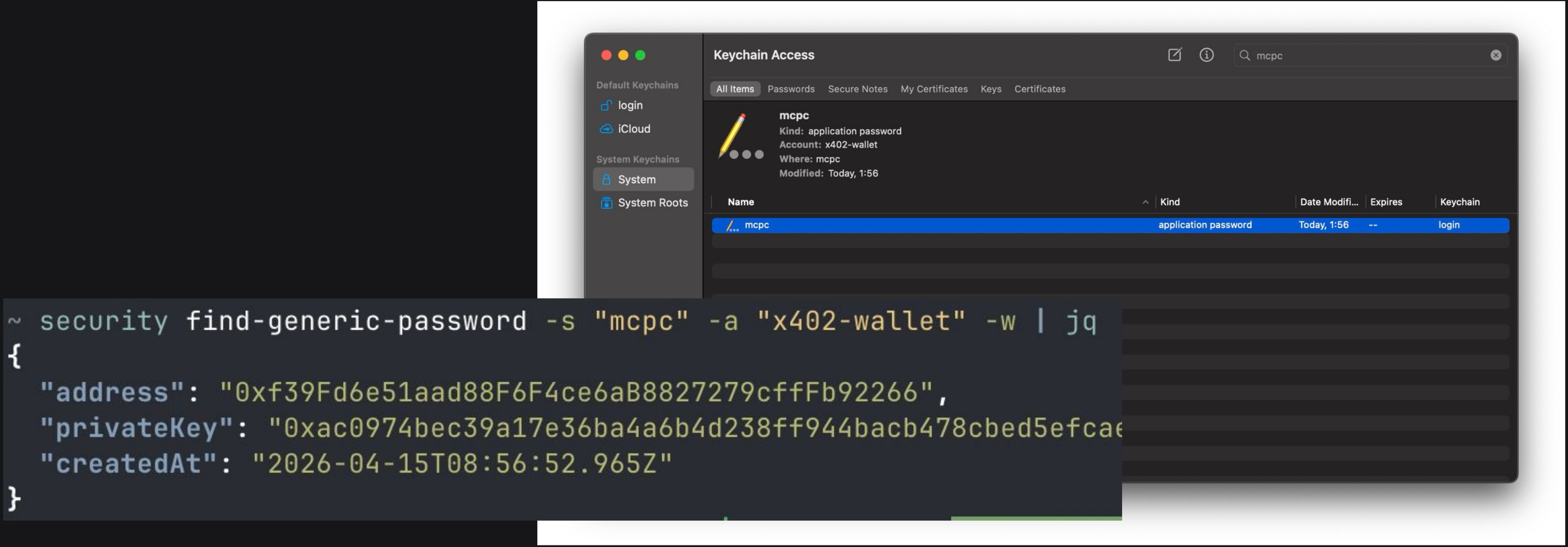
`mcpc x402`

```
~  mcpc x402 init
✓ Wallet created
i Address: 0x01382f6B1DFd156bfc8Fa522D9bC8e1D0D01C781
i Fund this address with USDC on Base to use x402 payments.

  Scan to fund this wallet:
```

Import the **wallet**

`mcpc x402 import`

- stored in the **system keychain** or in **~/.mcpc/wallets.json**



Keychain Access
mcpc
Default Keychains
login
iCloud
System Keychains
System
System Roots
All Items
Passwords
Secure Notes
My Certificates
Keys
Certificates

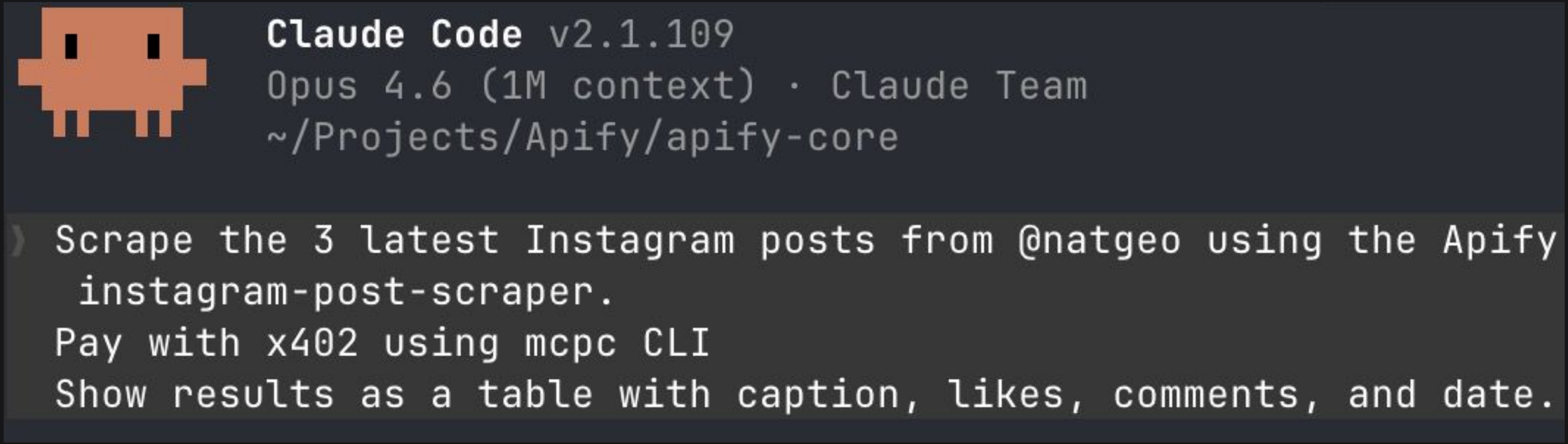
mcpc
Kind: application password
Account: x402-wallet
Where: mcpc
Modified: Today, 1:56

Name	Kind	Date Modifi...	Expires	Keychain
mcpc	application password	Today, 1:56	--	login

```
~ security find-generic-password -s "mcpc" -a "x402-wallet" -w | jq
{
  "address": "0xf39Fd6e51aad88F6F4ce6aB8827279cffFb92266",
  "privateKey": "0xac0974bec39a17e36ba4a6b4d238ff944bacb478cbed5efcae
  "createdAt": "2026-04-15T08:56:52.965Z"
}
```

Tell the **agent** to use `mcpc`

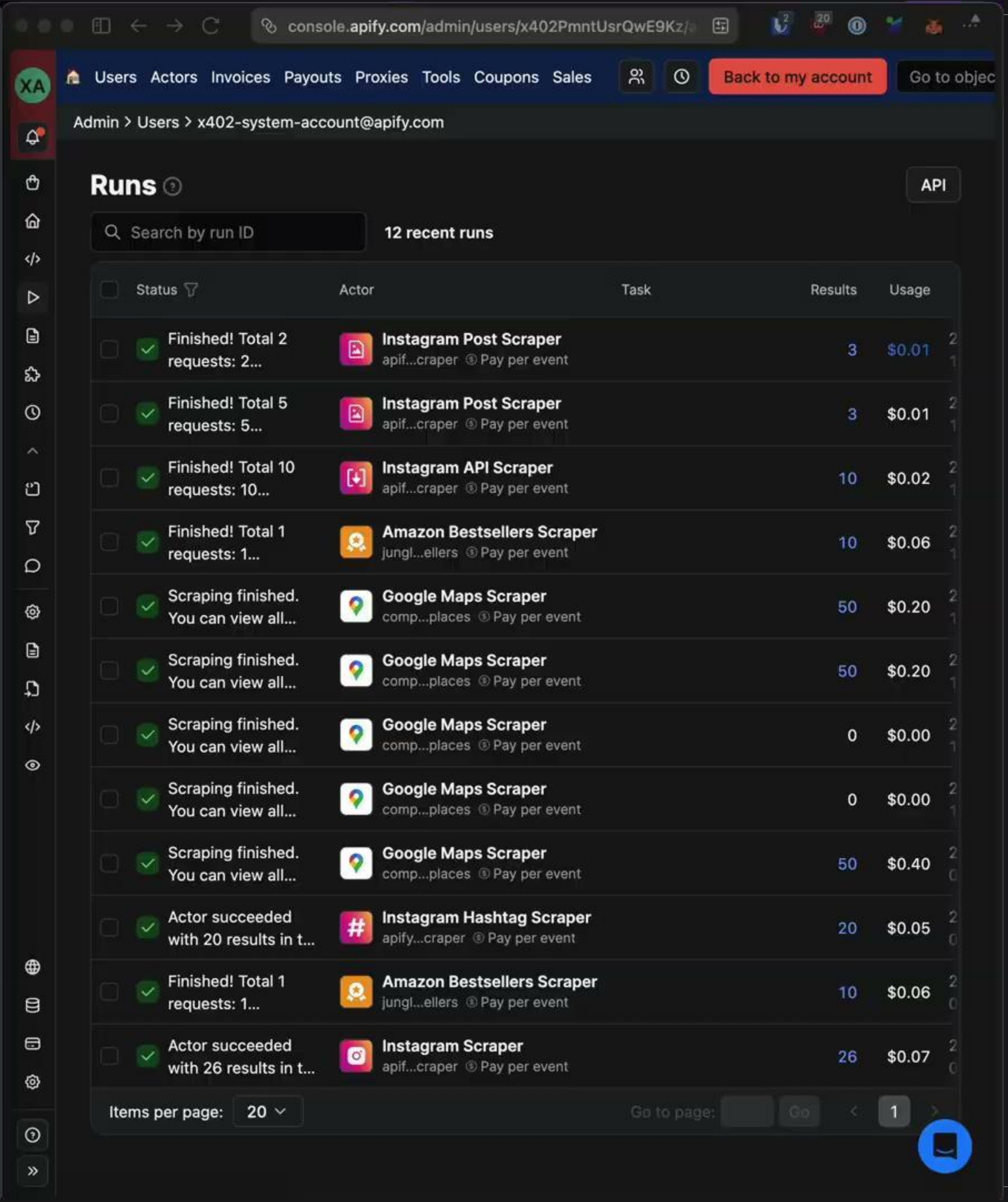
- fully local, your private key won't end up in the LLM context
- saves context, no need create the signing code



```
Claude Code v2.1.109
Opus 4.6 (1M context) · Claude Team
~/Projects/Apify/apify-core

❯ Scrape the 3 latest Instagram posts from @natgeo using the Apify
   instagram-post-scraper.
  Pay with x402 using mcpc CLI
  Show results as a table with caption, likes, comments, and date.
```

Demo:
MCP server and **mcpc**

```
~/Projects/demo
steps on ~/Projects/demo
>
```



console.apify.com/admin/users/x402PmntUsrQwE9Kz/
XA
Users
Actors
Invoices
Payouts
Proxies
Tools
Coupons
Sales
Back to my account
Go to objec

Admin > Users > x402-system-account@apify.com

Runs

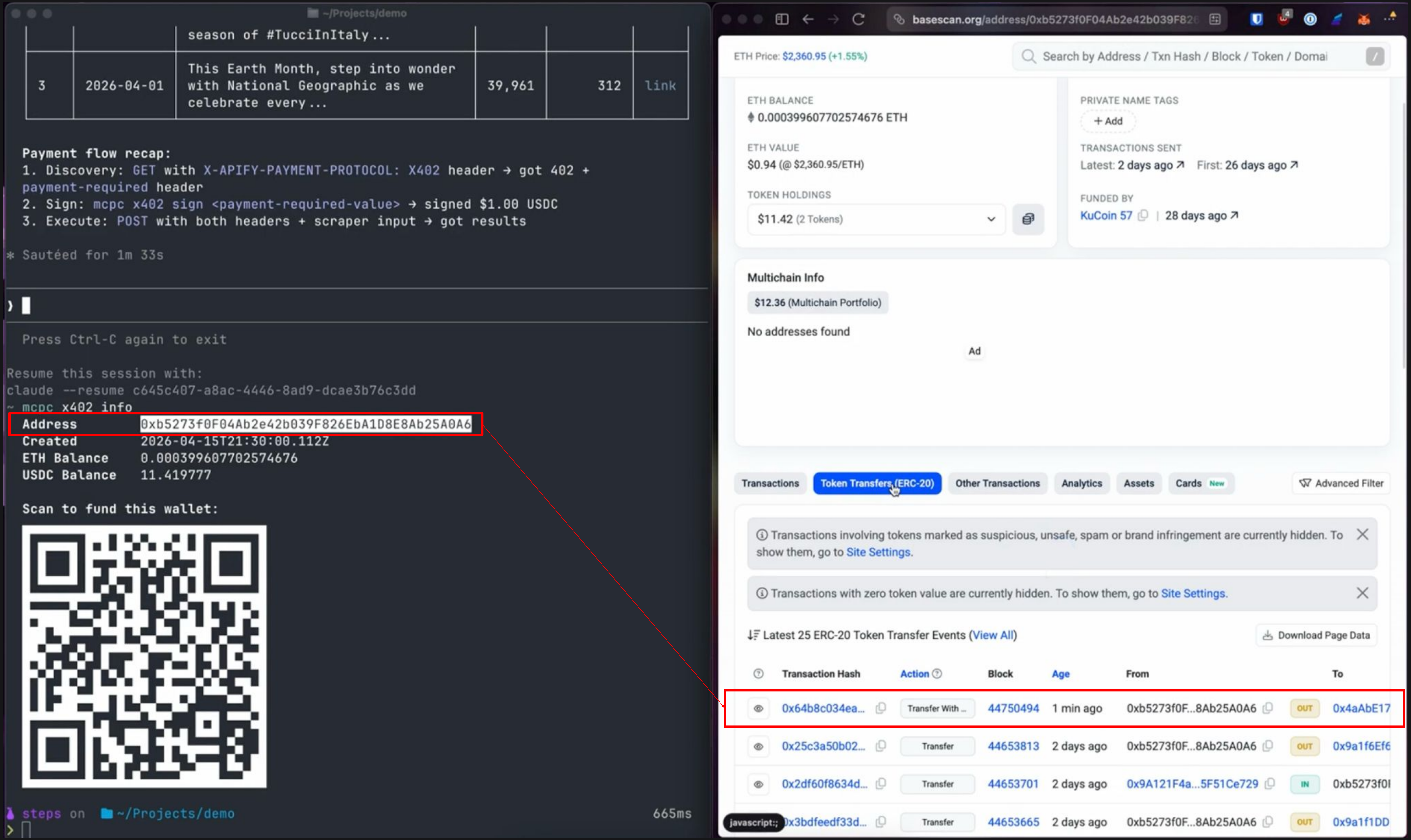
API
Search by run ID

12 recent runs

Status	Actor	Task	Results	Usage
Finished! Total 2 requests: 2...	Instagram Post Scraper apif...craper Pay per event		3	$0.01
Finished! Total 5 requests: 5...	Instagram Post Scraper apif...craper Pay per event		3	$0.01
Finished! Total 10 requests: 10...	Instagram API Scraper apif...craper Pay per event		10	$0.02
Finished! Total 1 requests: 1...	Amazon Bestsellers Scraper jungl...ellers Pay per event		10	$0.06
Scraping finished. You can view all...	Google Maps Scraper comp...places Pay per event		50	$0.20
Scraping finished. You can view all...	Google Maps Scraper comp...places Pay per event		50	$0.20
Scraping finished. You can view all...	Google Maps Scraper comp...places Pay per event		0	$0.00
Scraping finished. You can view all...	Google Maps Scraper comp...places Pay per event		0	$0.00
Scraping finished. You can view all...	Google Maps Scraper comp...places Pay per event		50	$0.40
Actor succeeded with 20 results in t...	Instagram Hashtag Scraper apify...craper Pay per event		20	$0.05
Finished! Total 1 requests: 1...	Amazon Bestsellers Scraper jungl...ellers Pay per event		10	$0.06
Actor succeeded with 26 results in t...	Instagram Scraper apif...craper Pay per event		26	$0.07

Items per page: 20

Go to page:
Go
1



~/Projects/demo

```
│   │            │ season of #TucciInItaly...        │        │       │      │
├───┼────────────┼───────────────────────────────────┼────────┼───────┼──────┤
│   │            │ This Earth Month, step into wonder│        │       │      │
│ 3 │ 2026-04-01 │ with National Geographic as we    │ 39,961 │   312 │ link │
│   │            │ celebrate every...                │        │       │      │
└───┴────────────┴───────────────────────────────────┴────────┴───────┴──────┘

  Payment flow recap:
  1. Discovery: GET with X-APIFY-PAYMENT-PROTOCOL: X402 header → got 402 +
  payment-required header
  2. Sign: mcpc x402 sign <payment-required-value> → signed $1.00 USDC
  3. Execute: POST with both headers + scraper input → got results

* Sautéed for 1m 33s

› █

  Press Ctrl-C again to exit

Resume this session with:
claude --resume c645c407-a8ac-4446-8ad9-dcae3b76c3dd
~ mcpc x402 info
  Address        0xb5273f0F04Ab2e42b039F826EbA1D8E8Ab25A0A6
  Created        2026-04-15T21:30:00.112Z
  ETH Balance    0.000399607702574676
  USDC Balance   11.419777

  Scan to fund this wallet:

steps on ~/Projects/demo                                                  665ms
›
```

basescan.org/address/0xb5273f0F04Ab2e42b039F82

ETH Price: $2,360.95 (+1.55%)

Search by Address / Txn Hash / Block / Token / Domai

ETH BALANCE
0.000399607702574676 ETH

ETH VALUE
$0.94 (@ $2,360.95/ETH)

TOKEN HOLDINGS
$11.42 (2 Tokens)

PRIVATE NAME TAGS

TRANSACTIONS SENT
Latest: 2 days ago ↗ First: 26 days ago ↗

FUNDED BY
KuCoin 57 | 28 days ago ↗

Multichain Info

$12.36 (Multichain Portfolio)

No addresses found

Ad
Transactions
Token Transfers (ERC-20)
Other Transactions
Analytics
Assets
Cards New
Advanced Filter

ⓘ Transactions involving tokens marked as suspicious, unsafe, spam or brand infringement are currently hidden. To show them, go to Site Settings.

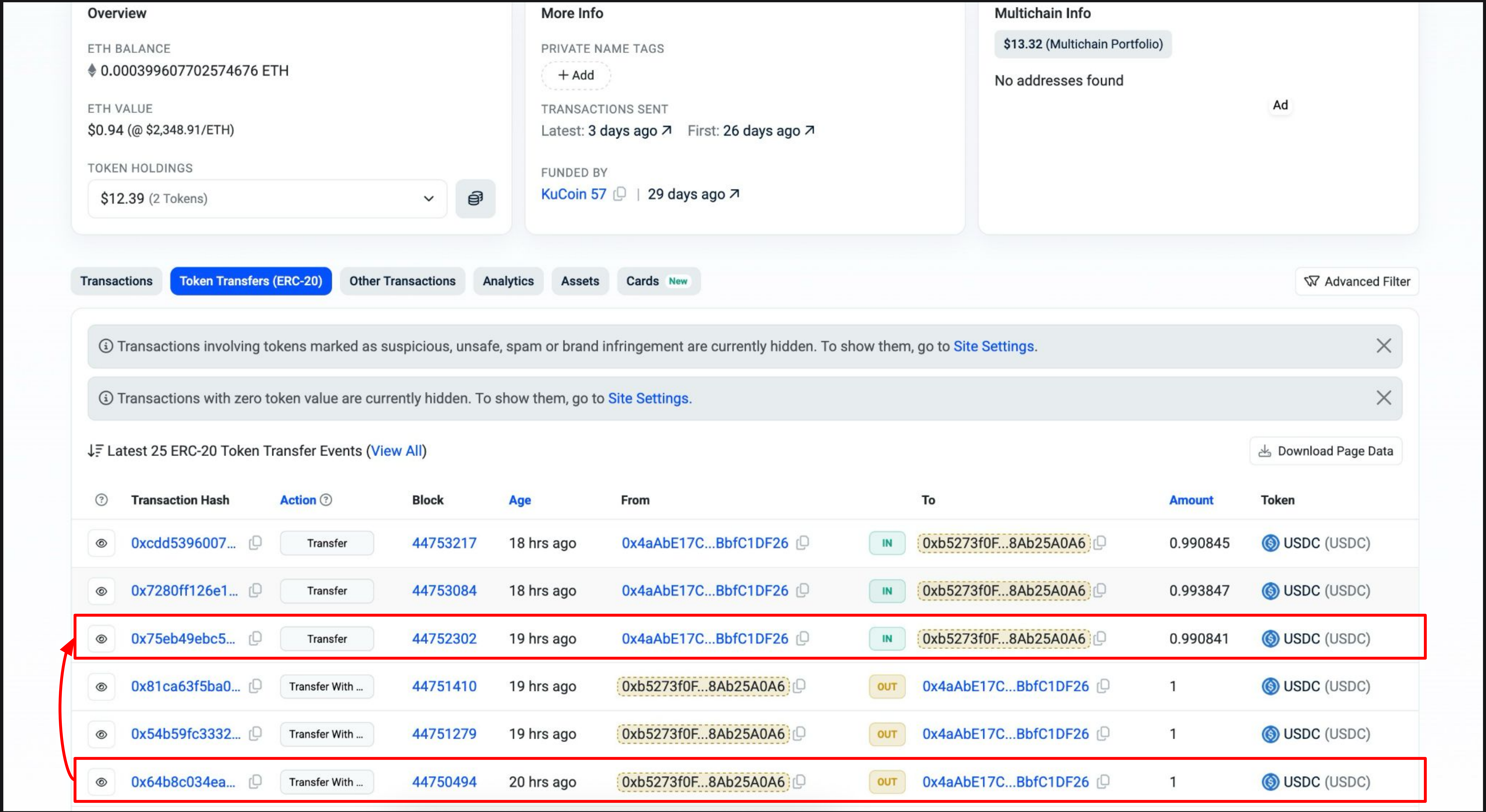
ⓘ Transactions with zero token value are currently hidden. To show them, go to Site Settings.

Latest 25 ERC-20 Token Transfer Events (View All)

Download Page Data

	Transaction Hash	Action	Block	Age	From		To
	0x64b8c034ea...	Transfer With ...	44750494	1 min ago	0xb5273f0F...8Ab25A0A6	OUT	0x4aAbE17
	0x25c3a50b02...	Transfer	44653813	2 days ago	0xb5273f0F...8Ab25A0A6	OUT	0x9a1f6Ef6
	0x2df60f8634d...	Transfer	44653701	2 days ago	0x9A121F4a...5F51Ce729	IN	0xb5273f0I
	0x3bdfeedf33d...	Transfer	44653665	2 days ago	0xb5273f0F...8Ab25A0A6	OUT	0x9a1f1DD

javascript:;



Overview

ETH BALANCE

0.000399607702574676 ETH

ETH VALUE

$0.94 (@ $2,348.91/ETH)

TOKEN HOLDINGS

$12.39 (2 Tokens)

More Info

PRIVATE NAME TAGS

TRANSACTIONS SENT

Latest: 3 days ago ↗ First: 26 days ago ↗

FUNDED BY

KuCoin 57 | 29 days ago ↗

Multichain Info

$13.32 (Multichain Portfolio)

No addresses found

Transactions involving tokens marked as suspicious, unsafe, spam or brand infringement are currently hidden. To show them, go to Site Settings.

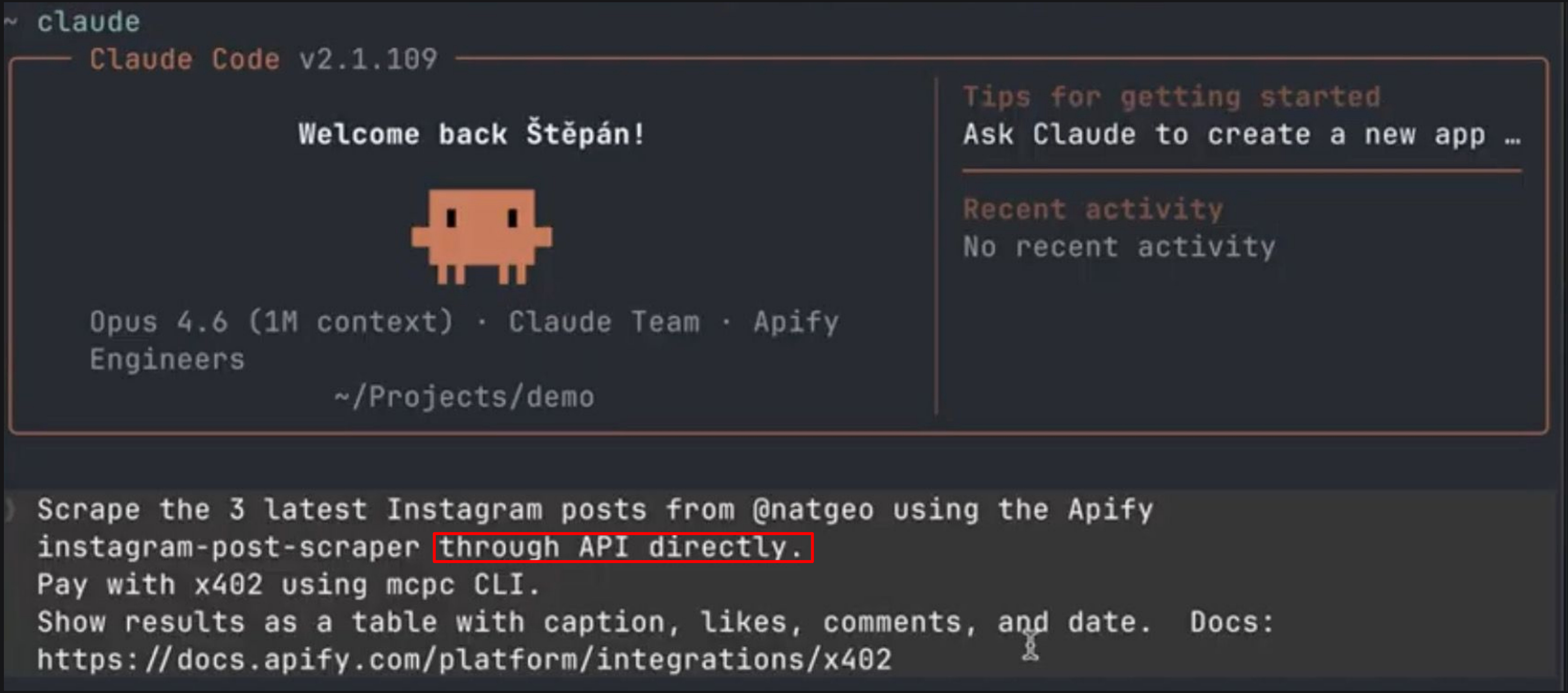
Transactions with zero token value are currently hidden. To show them, go to Site Settings.

Latest 25 ERC-20 Token Transfer Events (View All)

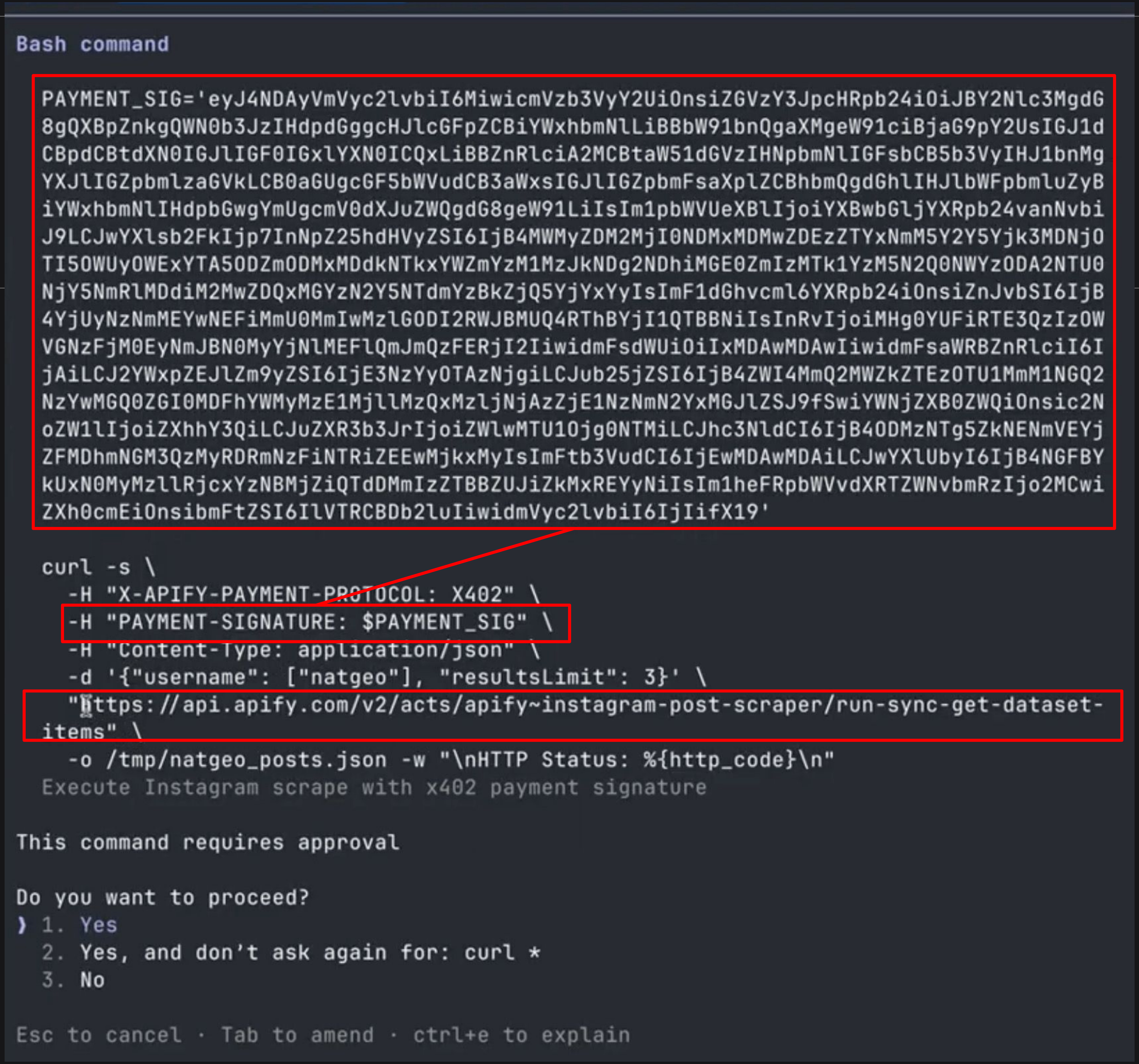
Transaction Hash	Action	Block	Age	From		To	Amount	Token
0xcdd5396007...	Transfer	44753217	18 hrs ago	0x4aAbE17C...BbfC1DF26	IN	0xb5273f0F...8Ab25A0A6	0.990845	USDC (USDC)
0x7280ff126e1...	Transfer	44753084	18 hrs ago	0x4aAbE17C...BbfC1DF26	IN	0xb5273f0F...8Ab25A0A6	0.993847	USDC (USDC)
0x75eb49ebc5...	Transfer	44752302	19 hrs ago	0x4aAbE17C...BbfC1DF26	IN	0xb5273f0F...8Ab25A0A6	0.990841	USDC (USDC)
0x81ca63f5ba0...	Transfer With ...	44751410	19 hrs ago	0xb5273f0F...8Ab25A0A6	OUT	0x4aAbE17C...BbfC1DF26	1	USDC (USDC)
0x54b59fc3332...	Transfer With ...	44751279	19 hrs ago	0xb5273f0F...8Ab25A0A6	OUT	0x4aAbE17C...BbfC1DF26	1	USDC (USDC)
0x64b8c034ea...	Transfer With ...	44750494	20 hrs ago	0xb5273f0F...8Ab25A0A6	OUT	0x4aAbE17C...BbfC1DF26	1	USDC (USDC)

Why **MCP**?

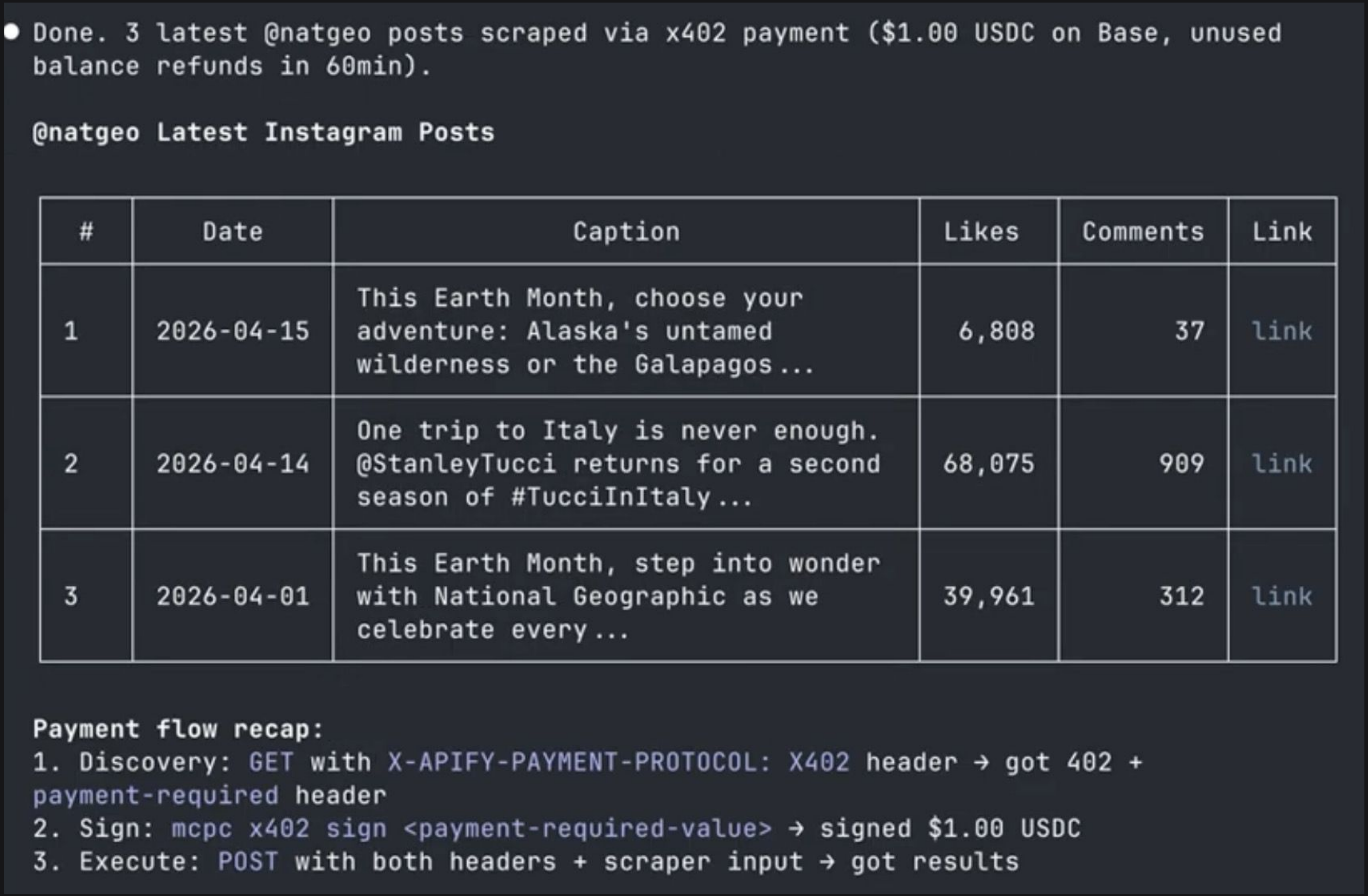
Rest **API** and **mcpc**



```
~ claude
╭─── Claude Code v2.1.109 ──────────────────────────────────────────────────────╮
│                                            │ Tips for getting started          │
│           Welcome back Štěpán!             │ Ask Claude to create a new app …  │
│                                            │ ───────────────────────────────── │
│                                            │ Recent activity                   │
│                                            │ No recent activity                │
│                                            │                                   │
│   Opus 4.6 (1M context) · Claude Team · Apify                                  │
│   Engineers                                │                                   │
│              ~/Projects/demo               │                                   │
╰────────────────────────────────────────────────────────────────────────────────╯

 Scrape the 3 latest Instagram posts from @natgeo using the Apify
 instagram-post-scraper through API directly.
 Pay with x402 using mcpc CLI.
 Show results as a table with caption, likes, comments, and date.  Docs:
 https://docs.apify.com/platform/integrations/x402
```



```
Bash command

  PAYMENT_SIG='eyJ4NDAyVmVyc2lvbiI6MiwicmVzb3VyY2UiOnsiZGVzY3JpcHRpb24iOiJBY2Nlc3MgdG
  8gQXBpZnkgQWN0b3JzIHdpdGggcHJlcGFpZCBiYWxhbmNlLiBBbW91bnQgaXMgeW91ciBjaG9pY2UsIGJ1d
  CBpdCBtdXN0IGJlIGF0IGxlYXN0ICQxLiBBZnRlciA2MCBtaW51dGVzIHNpbmNlIGFsbCB5b3VyIHJ1bnMg
  YXJlIGZpbmlzaGVkLCB0aGUgcGF5bWVudCB3aWxsIGJlIGZpbmFsaXplZCBhbmQgdGhlIHJlbWFpbmluZyB
  iYWxhbmNlIHdpbGwgYmUgcmV0dXJuZWQgdG8geW91LiIsIm1pbWVUeXBlIjoiYXBwbGljYXRpb24vanNvbi
  J9LCJwYXlsb2FkIjp7InNpZ25hdHVyZSI6IjB4MWMyZDM2MjI0NDMxMDMwZDEzZTYxNmM5Y2Y5Yjk3MDNjO
  TI5OWUyOWExYTA5ODZmODMxMDdkNTkxYWZmYzM1MzJkNDg2NDhiMGE0ZmIzMTk1YzM5N2Q0NWYzODA2NTU0
  NjY5NmRlMDdiM2MwZDQxMGYzN2Y5NTdmYzBkZjQ5YjYxYyIsImF1dGhvcml6YXRpb24iOnsiZnJvbSI6IjB
  4YjUyNzNmMEYwNEFiMmU0MmIwMzlGODI2RWJBMUQ4RThBYjI1QTBBNiIsInRvIjoiMHg0YUFiRTE3QzIzOW
  VGNzFjM0EyNmJBN0MyYjNlMEFlQmJmQzFERjI2IiwidmFsdWUiOiIxMDAwMDAwIiwidmFsaWRBZnRlciI6I
  jAiLCJ2YWxpZEJlZm9yZSI6IjE3NzYyOTAzNjgiLCJub25jZSI6IjB4ZWI4MmQ2MWZkZTEzOTU1MmM1NGQ2
  NzYwMGQ0ZGI0MDFhYWMyMzE1MjllMzQxMzljNjAzZjE1NzNmN2YxMGJlZSJ9fSwiYWNjZXB0ZWQiOnsic2N
  oZW1lIjoiZXhhY3QiLCJuZXR3b3JrIjoiZWlwMTU1Ojg0NTMiLCJhc3NldCI6IjB4ODMzNTg5ZkNENmVEYj
  ZFMDhmNGM3QzMyRDRmNzFiNTRiZEEwMjkxMyIsImFtb3VudCI6IjEwMDAwMDAiLCJwYXlUbyI6IjB4NGFBY
  kUxN0MyMzllRjcxYzNBMjZiQTdDMmIzZTBBZUJiZkMxREYyNiIsIm1heFRpbWVvdXRTZWNvbmRzIjo2MCwi
  ZXh0cmEiOnsibmFtZSI6IlVTRCBDb2luIiwidmVyc2lvbiI6IjIifX19'

  curl -s \
    -H "X-APIFY-PAYMENT-PROTOCOL: X402" \
    -H "PAYMENT-SIGNATURE: $PAYMENT_SIG" \
    -H "Content-Type: application/json" \
    -d '{"username": ["natgeo"], "resultsLimit": 3}' \
    "https://api.apify.com/v2/acts/apify~instagram-post-scraper/run-sync-get-dataset-
  items" \
    -o /tmp/natgeo_posts.json -w "\nHTTP Status: %{http_code}\n"
  Execute Instagram scrape with x402 payment signature

This command requires approval

Do you want to proceed?
❱ 1. Yes
  2. Yes, and don't ask again for: curl *
  3. No

Esc to cancel · Tab to amend · ctrl+e to explain
```



● Done. 3 latest @natgeo posts scraped via x402 payment ($1.00 USDC on Base, unused balance refunds in 60min).

@natgeo Latest Instagram Posts

#	Date	Caption	Likes	Comments	Link
1	2026-04-15	This Earth Month, choose your adventure: Alaska's untamed wilderness or the Galapagos ...	6,808	37	link
2	2026-04-14	One trip to Italy is never enough. @StanleyTucci returns for a second season of #TucciInItaly ...	68,075	909	link
3	2026-04-01	This Earth Month, step into wonder with National Geographic as we celebrate every ...	39,961	312	link

Payment flow recap:
1. Discovery: `GET` with `X-APIFY-PAYMENT-PROTOCOL: X402` header → got 402 + `payment-required` header
2. Sign: `mcpc x402 sign <payment-required-value>` → signed $1.00 USDC
3. Execute: `POST` with both headers + scraper input → got results

Demo:
Rest **API** and **mcpc**

apify

Stepan Skopek

@stepskop

Apify docs for x402

github.com/apify/mcpc